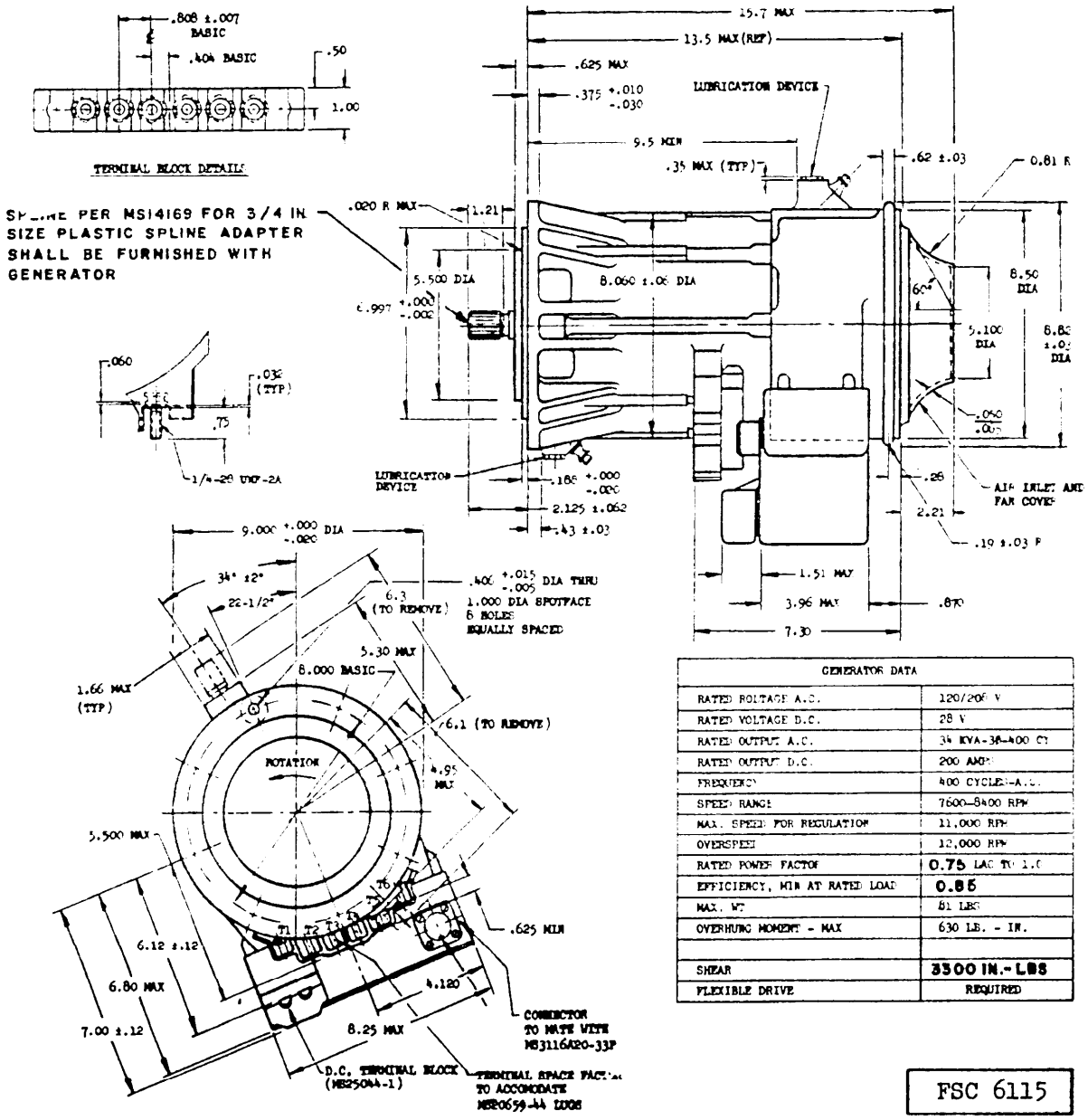
MILITARY SPECIFICATION SHEET

GENERATOR SYSTEM, 34 KVA, 400 CYCLE, ALTERNATING CURRENT, INTEGRALLY EXCITED, BRUSHLESS TYPE AND 200 AMPS AT 28 VOLTS, D. C.

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the generator described herein shall consist of this document and the issue in effect of MIL-G-21480.



GENERATOR DATA	
RATED VOLTAGE A.C.	120/208 V
RATED VOLTAGE D.C.	28 V
RATED OUTPUT A.C.	34 KVA-3Φ-400 CY
RATED OUTPUT D.C.	200 AMP
FREQUENCY	400 CYCLES-A.C.
SPEED RANGE	7600-8400 RPM
MAX. SPEED FOR REGULATION	11,000 RPM
OVERSPEED	12,000 RPM
RATED POWER FACTOR	0.75 LAG TO 1.0
EFFICIENCY, MIN AT RATED LOAD	0.85
MAX. WT	81 LBS
OVERHUNG MOMENT - MAX	630 LB. - IN.
SHEAR	3500 IN.-LBS
FLEXIBLE DRIVE	REQUIRED

FSC 6115

FIGURE 1. M21480/8-5, GENERATOR

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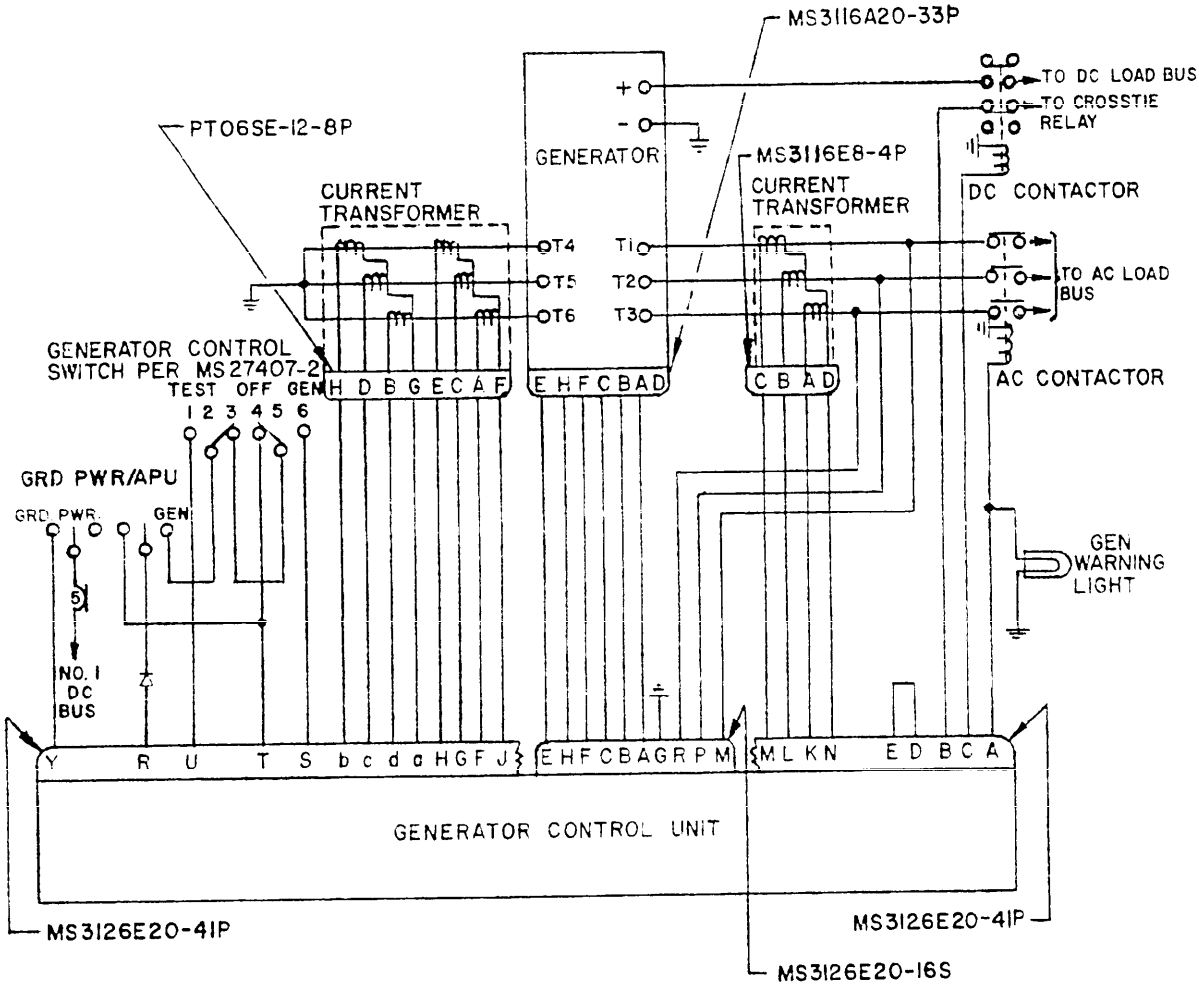


FIGURE 2. TEST CIRCUIT

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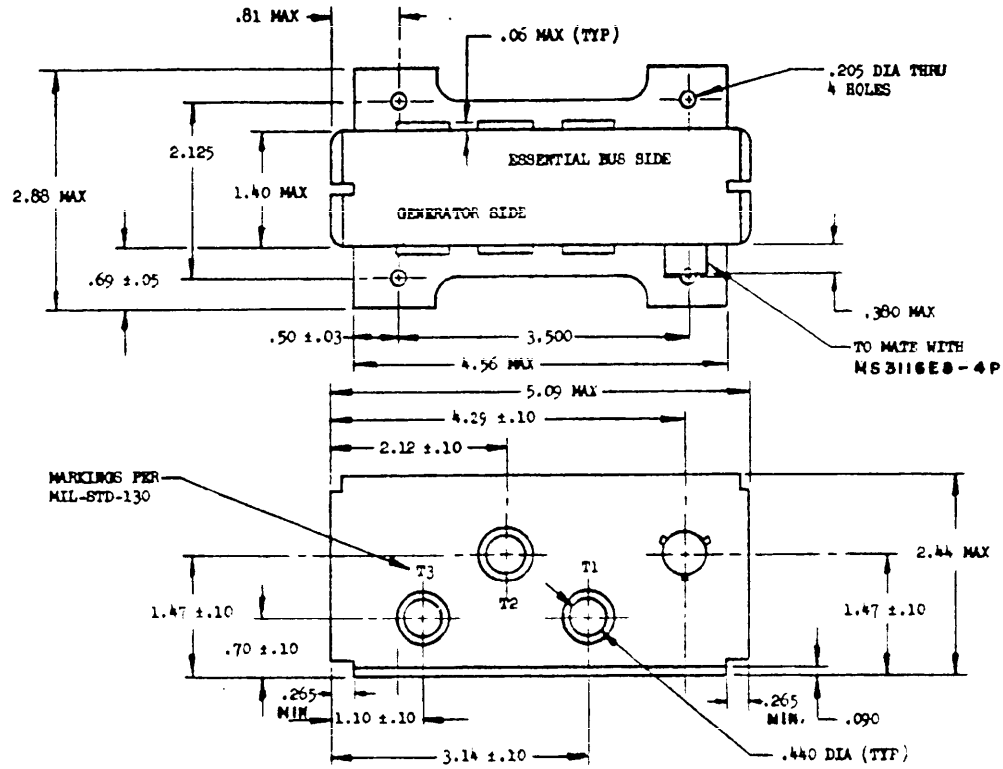


FIGURE 3. M21480/8-2, CURRENT TRANSFORMER, WT.- 1.3 LB. MAX.

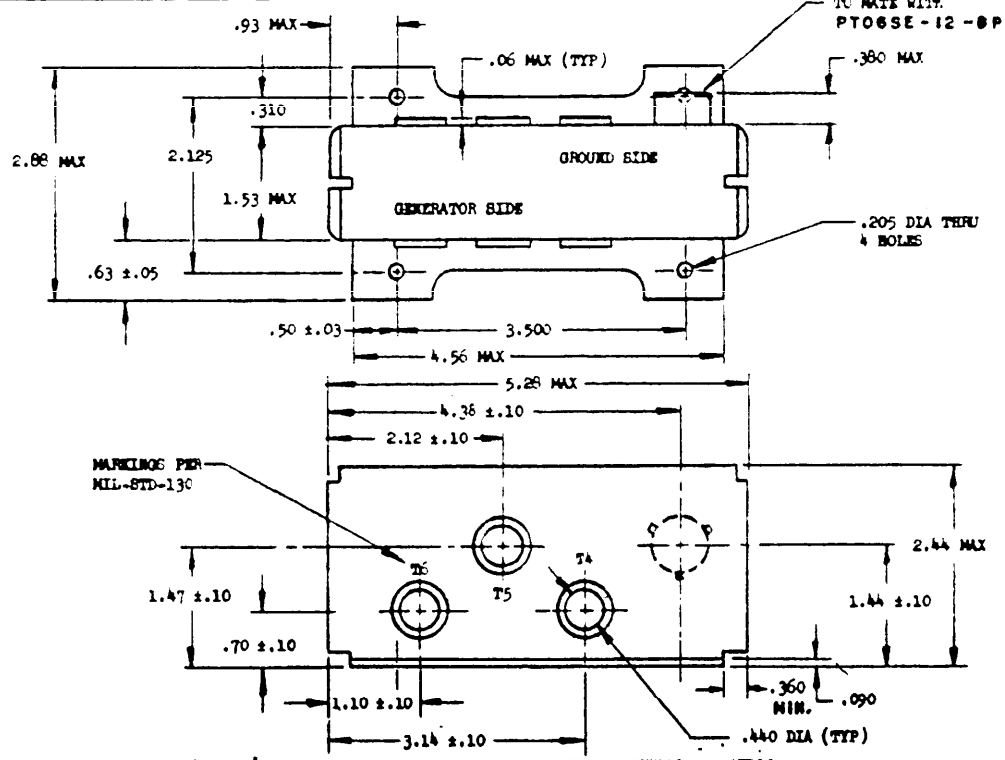


FIGURE 4. M21480/8-3, CURRENT TRANSFORMER, WT.- 1.5 LB. MAX.

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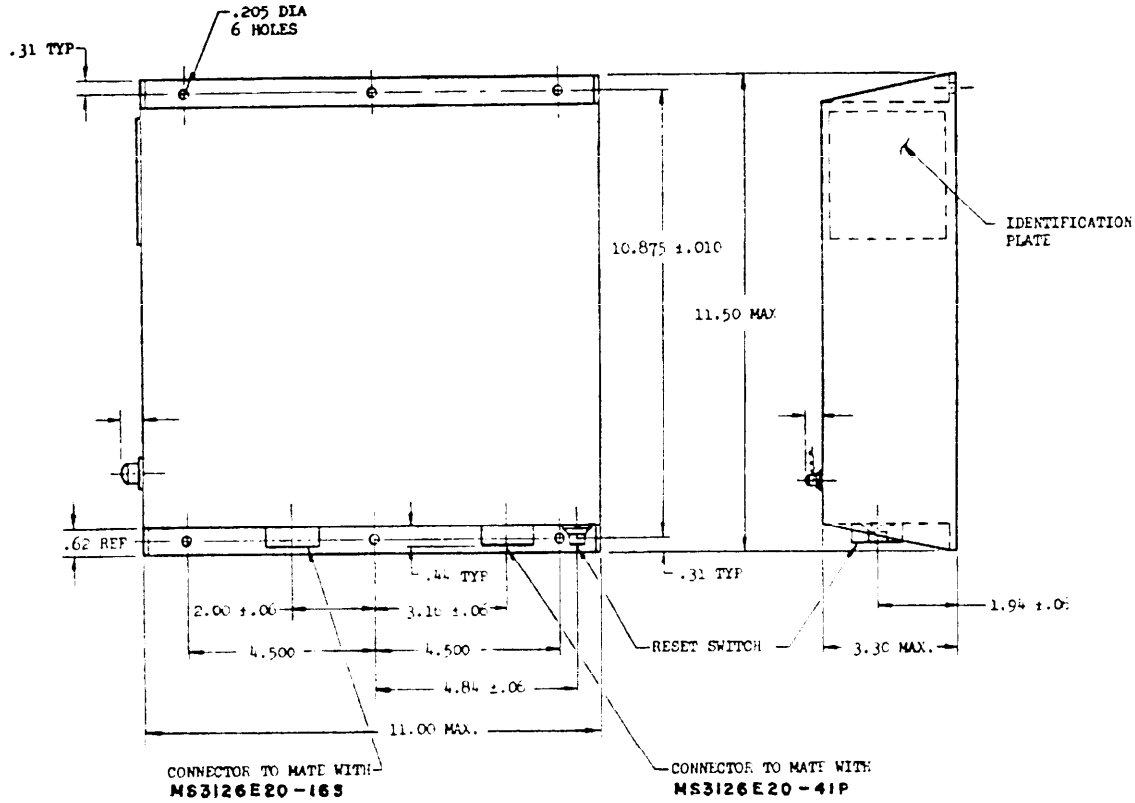


FIGURE 5. M21480/8-4, GENERATOR CONTROL UNIT. WT.- 10.5 LB. MAX.

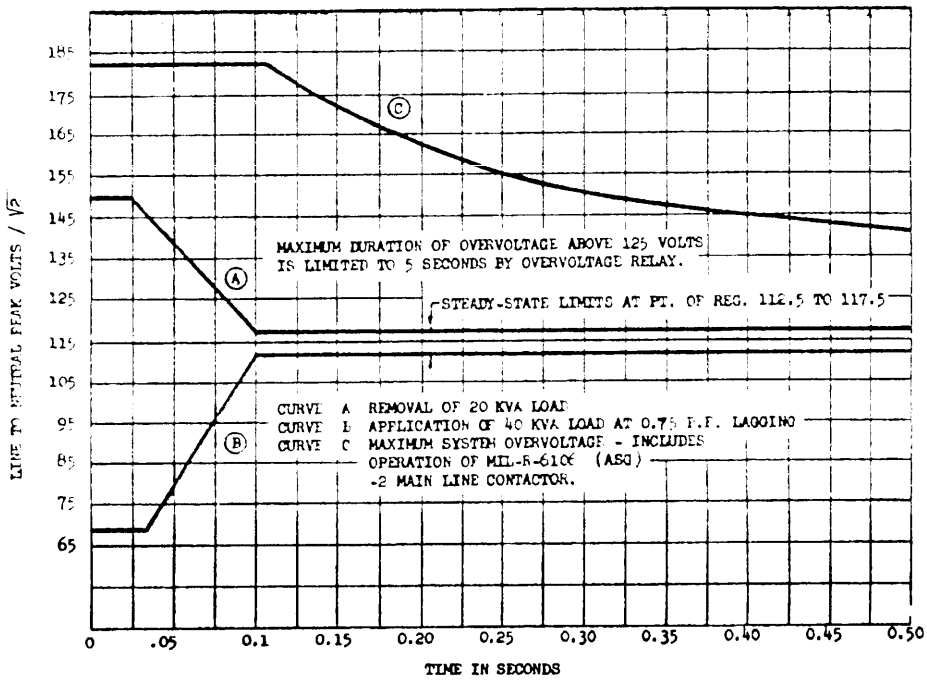


FIGURE 6. TRANSIENT, STEADY STATE AND OVERVOLTAGE LIMITS

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APPROXIMATE TOTAL KVA AC + DC	LOAD		RPM MIN	AMBIENT TEMP MAX	ALTITUDE
	KVA AC	AMPS DC			
20	14	200	6800	71°C	SEA LEVEL
20	20	20	6800	71°C	SEA LEVEL
20	14	200	6800	0°C	15,000 FEET
20	20	20	6800	0°C	15,000 FEET
40	34	200	7200	25°C	SEA LEVEL
36	30	200	7200	71°C	SEA LEVEL
40	34	200	7600	71°C	SEA LEVEL
40	34	200	7600	0°C	15,000 FEET

ALL AC LOADS AT 0.75 LAG POWER FACTOR

FIGURE 7. RATING

APPROXIMATE TOTAL KV AC + DC	LOAD		RPM MIN	AMBIENT TEMP MAX	ALTITUDE	TIME
	KVA AC	AMPS DC				
30	24	200	6800	71°C	SEA LEVEL	5 SECONDS
30	24	200	6800	0°C	15,000 FEET	5 SECONDS
30	24	200	6800	-20°C	25,000 FEET	5 SECONDS
30	30	20	6800	71°C	SEA LEVEL	5 SECONDS
30	30	20	6800	0°C	15,000 FEET	5 SECONDS
30	30	20	6800	-20°C	25,000 FEET	5 SECONDS
48	42	200	7200	25°C	SEA LEVEL	2 MINUTES
52	46	200	7200	25°C	SEA LEVEL	5 SECONDS
60	54	200	8000	25°C	SEA LEVEL	2 MINUTES
80	74	200	8000	25°C	SEA LEVEL	5 SECONDS
52	46	200	7600	25°C	SEA LEVEL	2 MINUTES
60	54	200	7600	71°C	SEA LEVEL	5 SECONDS

ALL ABOVE OVERLOADS WILL BE APPLIED AFTER RUNNING THE UNIT AT 25 KVA TOTAL LOAD (19 KVA A-C + 200 AMPS D-C) AT 7200 RPM STEADY-STATE CONTINUOUS

FIGURE 8. OVERLOADS

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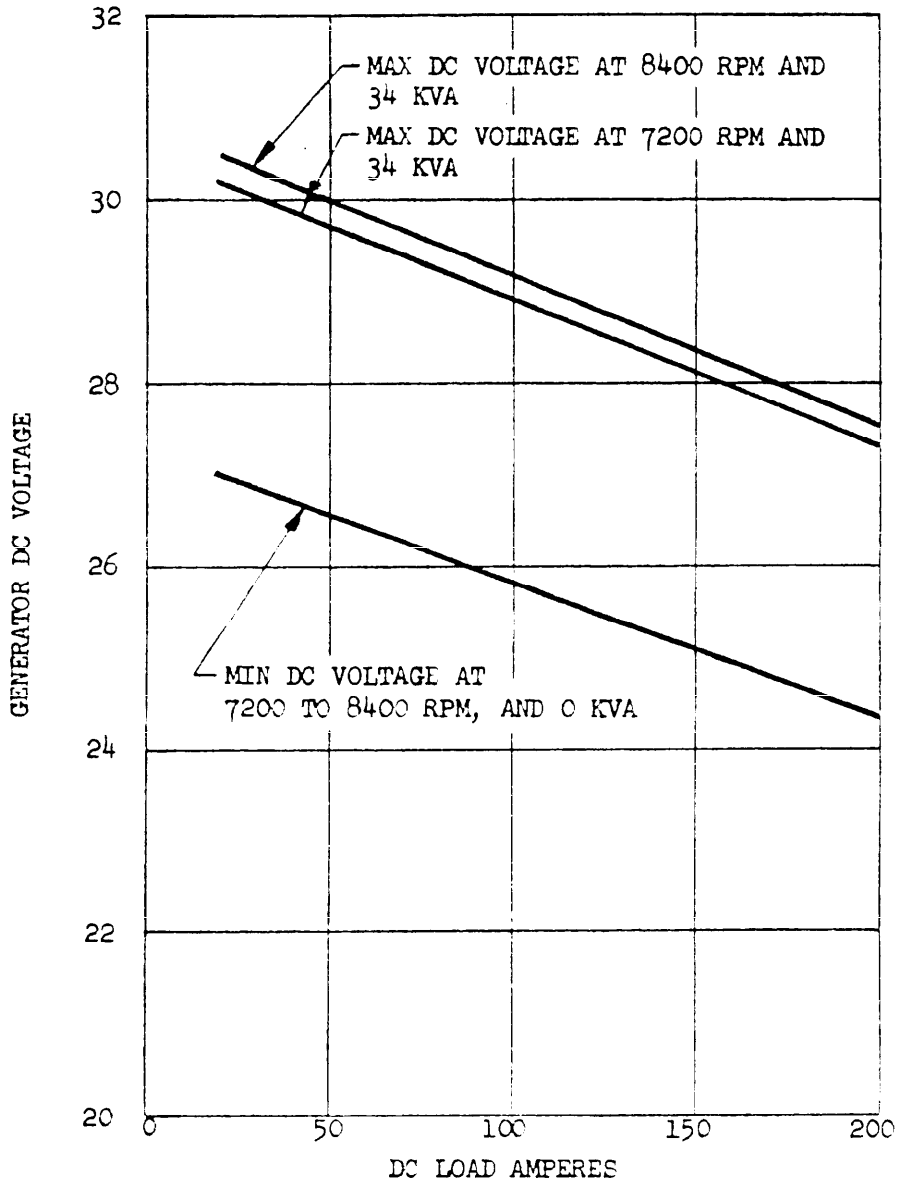


FIGURE 9. DC VOLTAGE LIMITS

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REQUIREMENTS:

The requirements of MIL-G-21480 apply with the following exceptions to existing paragraphs:

Delete the following paragraphs:

3.4.9.6.2, 3.5.7.5, 4.4.1.2.2, 4.4.2.3.1, 4.4.2.3.2, 4.5.20.2

3.3.3.1 Add: "No magnesium shall be used in the generator."

3.4.2.3 Change "Specification MIL-E-5272, Procedure I" to "MIL-E-81910, paragraph 4.6".

3.4.2.4 Change "Specification MIL-E-5272, Procedure I" to "MIL-E-81910, paragraph 4.3".

3.4.3 Add: "The generator in this application normally will operate continuously with shaft vertical and mounting flange up."

3.4.8 Delete first sentence and add: "The generator shall be an air-cooled, brushless AC/DC machine, three phase, wye-connected, 400Hz, 120/208 volt AC, and 200 amps at 28 volts DC." Change "Figure 3" to "Figure 6" of this specification.

3.4.8.1 Delete first sentence and add: "The system shall be capable of continuously delivering its rated capacity specified in Figure 7 of this specification."

3.4.8.2 Delete and add: "The system shall be capable of delivering the overloads as specified in Figure 8 of this specification."

3.4.8.3 Delete and add: "The AC system shall be capable of supplying a minimum of 225 amperes single phase line-to-line and line-to-neutral, and 180 amperes three phase short circuit current for 5 seconds at 8,000 rpm when the short circuits are applied to the AC bus. The DC system shall be capable of withstanding a short circuit applied to the DC bus for 5 seconds without causing damage to any system component."

3.4.9.2 Add: "Shear strength shall be 3,000 to 3,500 inch-pounds."

3.4.9.2.1 Grease lubricated bearings - When grease lubricated bearings are used, they shall be packed with grease meeting specification MIL-G-81322.

3.4.9.6.1 Add: "An integral fan shall be provided for self-cooling during all conditions specified herein."

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REQUIREMENTS (cont.)

3.4.9.6.1.1 Change "10,000 hour life" to "1,000 hour life".

3.4.9.7 Delete and add: "The generator, when operating within its rating and under the environmental conditions of this application, shall operate 1,000 hours between overhaul. The useful life shall be 5,000 hours or more. The addition of lubricant (MIL-G-81322) to the bearings shall be at intervals of not less than 125 hours. The addition of lubricant shall not require removal of generator."

3.4.10.1 Add: "The regulator voltage setting shall be preset at 115 ±1 volts at 25 ±5°C, 400 Hz, full rated load, P.F.=1.0."

3.4.10.3 Delete and add: "The accessory components, when operated within the temperature - altitude conditions of Figure 7, shall have a useful life of 5,000 hours or more."

3.4.11 Change "All system components" to "Generator and GCU."

3.4.13 Add: "The nomenclature shall be as follows:

Generator

AC - DC Generator
Rated Voltage AC: 120/208
Rated Voltage DC: 28
Rated Output AC: 34 KVA
Phase: 3
Frequency: 380/420 Hz
P.F.: 0.75 to 1.0
Rated Output DC: 200 AMPS
Speed Range: 7,600 - 8,400 RPM
Manufacturer's Part Number
Manufacturer's Serial Number
Military Part Number: M21480/8-5
Date of Manufacture
Manufacturer's Name or Trademark
Federal Stock Number
Contract Number
U.S.

Generator Control Unit

115/200 volts AC
3 Phase
400 Hz

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REQUIREMENTS, 3.4.13, Generator Control Unit (cont.)

Manufacturer's Part Number
Manufacturer's Serial Number
Military Part Number: M21480/8-4
Contract Number
Date of Manufacture
Manufacturer's Name or Trademark
Federal Stock Number
U.S.

Current Transformer

Manufacturer's Part Number
Manufacturer's Serial Number
Military Part Number M21480/8-(X)
Contract Number
Date of Manufacture
Manufacturer's Name or Trademark
Federal Stock Number
U.S."

X = Appropriate Transformer (Loadside) or (Groundside)

3.5.1 Delete and Add: "The transient and steady state AC voltage shall remain within the limits of Figure 6 for all balanced load conditions while the system is operating under all the environmental and load conditions defined in Figure 7. Steady state DC voltage shall remain within the limits of Figure 9."

3.5.2.1 Change "1.0 volt" to "3.3 volts".

3.5.2.2 Delete and add: "Frequency characteristics for the voltage modulation shall not exceed the following limits:

Frequency	Voltage Modulation (Peak to Peak)
1.0	0.5
1.7	0.5
10.0	3.3
25.0	3.3
65.0	0.5
100.0	0.5"

3.5.4 Delete and add: "The phase voltage wave form shall be within the following limits at any DC load from 20 to 200 amperes with AC loads between 3.5 KVA and 34 KVA:

(1) Crest Factor 1.41 ± 10%
(2) Any single harmonic not to exceed 3% of the fundamental"

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REQUIREMENTS (cont.)

3.5.7.2 Delete all after "defined by" and add: "Figure 6, Curve C of this specification sheet".

3.5.7.3 Delete and add: "The system shall provide for the generator to be disconnected from the load bus when any one phase voltage drops below 90 volts for a time delay of 5 seconds. The system shall be capable of being automatically reset when all three phase voltages exceed 105 volts."

3.5.7.4 Delete and add: "The system shall provide for the generator to be disconnected from the load bus in 3 to 7 seconds when an underfrequency of between 320 and 340 Hz exists. An underfrequency below 320 Hz shall cause the generator to be disconnected from the bus instantaneously. On increasing frequency, the generator shall be reconnected to the bus between 350 and 370 Hz."

3.5.7.6 Change "exceeds 25 amperes" to "is 20 $\pm$ 4 amperes".

Add new paragraph 3.5.7.8.

"3.5.7.8 Voltage limits - No fault on the system shall cause the AC voltage of any phase of the generating system to exceed the limits of Figure 6 of this specification sheet, nor the DC voltage to exceed the limits of Figure 9 of this specification sheet."

3.5.8 Delete and add: "The system shall meet the requirements of MIL-E-81910, Class No. III B for the generators and control equipment, except the ten microfarad feed-through capacitor shall be removed."

4.2.1.1 Add: "Qualification tests shall be supplemented with a minimum 100 hour failure free flight test before consideration is given to incorporating the item on the Qualified Products List."

4.4.1.1.1 Change "horizontal" to "vertical".

4.4.1.2.1 Delete last sentence.

4.4.2.1.1 Change "horizontal" to "vertical".

4.4.3.3 Add: "The point of regulation shall be located 40 feet from the generator terminals, four No. 4 copper wires shall serve as the three phase AC conductors, and two sets of two conductor No. 2 copper wires shall serve as the DC conductors."

4.4.3.7 Add: "Speed range for rated load is 6,800 to 9,200 RPM."

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REQUIREMENTS (cont.)

4.5.3 Change "Figure 3" to "Figure 6" of this specification.

4.5.9 Delete all after (d) and add:

"(d) At minimum, maximum, and average rated generator RPM.

The efficiency at full rated load shall not be less than 85%."

4.5.11 Change "1.0 volt" to "3.3 volts".

4.5.12 Delete and add: "The generator, generator control unit, and CT shall be subjected to a electromagnetic interference test in accordance with MIL-E-81910, for Class III B equipment."

4.5.13 Delete and add: "The crest factor and harmonic content, line-to-line, and line-to-neutral, of the output voltage shall be determined under the following conditions:

(a) At generator speed of 8,000 RPM for no load, and in 20% load increments to full rated load, unity power factor.

(b) Condition (a) shall be repeated for 0.75 power factor.

(c) Conditions (a) and (b) shall be repeated at generator speeds of 7,600 and 8,400 RPM.

The following requirements must be met during all above tests except no load. The line-to-neutral crest factor shall be 1.41 $\pm$ 10%. No harmonic shall exceed 3% of the fundamental."

4.5.15 Change "vertically" to "horizontally".

4.5.20.1 Delete and add: "The GCU and CT's shall be subjected to a Salt Fog Test in accordance with MIL-E-81910, paragraph 4.6. At the completion of this test, the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of full and 200% rated loads, and protective functions."

4.5.21 Delete and add: "The generator, GCU and CT's shall be subjected to a Fungus Test in accordance with MIL-E-81910, paragraph 4.3 with the installation instructions and mounting nuts (as required by paragraph 4.3.1) attached. At the completion of this test, the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of full and 200% rated load, and protective functions. Mounting nuts are to be unaffected and the package holding the nuts suitable for shipment. Instructions are to be legible."

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REQUIREMENTS (cont.)

4.5.22.1 Delete and add: "The generator and GCU shall be subjected to a Dust Test in accordance with MIL-E-81910, paragraph 4.2. At the completion of this test, the unit shall successfully demonstrate its buildup characteristics, the effects of application and removal of full and 200% rated loads, and protective functions."

4.5.23.1.1 Delete and add: "The generator shall be subjected to a Vibration Test in accordance with MIL-E-81910, paragraph 4.9, except that the vibration range shall be from 5 to 500 Hz and the amplitude of vibration monitored at the anti-drive end of the generator during the main bending mode shall be limited to 20g's but the vibration input shall not be lowered below 5g's. At the completion of this test, the unit shall successfully demonstrate its buildup characteristics, and the effect of application and removal of full and 200% rated loads."

4.5.24 Delete and add: "The generator, GCU, and CT's shall be subjected to a Shock Test in accordance with MIL-E-81910, paragraph 4.7. At the completion of Procedure I, the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of full and 200% rated loads, and protective functions. There shall be no failure of the mounting attachments, and the test item shall remain in place and not create hazard during Procedure III."

4.5.25 Delete and add: "The generator, GCU, and CT's shall be subjected to a Humidity Test in accordance with MIL-E-81910, paragraph 4.4. At the completion of this test, the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of full and 200% rated loads, and protective functions."

4.5.27 Add: "Failure shall occur within the torque limits specified in paragraph 3.4.9.2 of the specification sheet."

4.5.30 Delete first sentence and add: "All system components except the generator shall be subjected to the acceleration tests of MIL-E-81910. At the completion of this test, the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of full and 200% rated loads, and protective functions."

NOTES:

1. Dimensions in inches. Unless otherwise specified, tolerances: decimals $\pm$.005, angles $\pm 1^{0}$.

2. Where only maximum dimensions are shown, the part need not have the shape shown, but should be contained within the outline shown.

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NOTES (cont.)

3. Protrusions such as rivet and screw heads shall be contained within the outline shown.

ITEM	PART NUMBER
Complete System (consists of one generator, one generator control unit and two three phase current transformers)	M21480/8-40
Generator	M21480/8-5
Generator Control Unit	M21480/8-4
Current Transformer (Load Side)	M21480/8-2
Current Transformer (Ground Side)	M21480/8-3

Revision letters are not used to denote changes due to the extensiveness of the changes.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-G-21480/8A(AS) GENERATOR SYSTEM

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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A. GIVE PARAGRAPH NUMBER AND WORDING

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C. REASON FOR RECOMMENDED CHANGE(S)

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